

Hiro Xiangyuan

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EDUCATION

The University of California, Berkeley | B.S in Data Science

Aug. 2023 – Present

Relevant Coursework: Data Structures & Algorithms, Principles & Techniques of Data Science, Computational Structures in Data Science, Probability for Data Science, Linear Algebra, Calculus, Deep Learning

WORK EXPERIENCE

Credo | *Software Engineering Intern*

May. 2026 – Present

- Implementing Python workflow automation engine using object-oriented design and state machine architecture with 6 node types, helping teams coordinate cross-department processes for faster product launches
- Building Django web application with MongoDB and MongoEngine to store workflow definitions, process instances, and audit logs, developing automation tools that help engineers track requests and meet deadlines efficiently

Whitney Laboratory | *Undergraduate Research Assistant*

Sep. 2025 – Present

- Built an automated data preprocessing pipeline standardizing pupil measurements for 200+ participants across 1,000+ trials using pandas z-score normalization, reducing inter-subject variability by 67%
- Developed linear regression analysis quantifying pupil-accuracy relationships ($R^2=0.43$, $p<0.001$) across 3,000+ trials, revealing 23% higher task performance during high-attention periods and validating engagement hypotheses

Lamu | *Data Engineer Intern*

Jan. 2026 – May. 2026

- Designed AWS cloud infrastructure (Lambda, RDS PostgreSQL, QuickSight) tracking 15+ KPIs and building automated data analytics dashboards, improving registration rates across distributed services
- Engineered ETL pipeline processing 500K+ user records from DynamoDB to PostgreSQL, writing SQL queries to track cohort retention and user growth trends that guided app and website improvements

BLCK UNICRN | *Software Engineer Intern*

Sep. 2025 - Dec. 2025

- Built Python ETL pipelines processing 50,000+ prompt-response pairs to train company chatbot datasets, achieving 95% data quality through automated validation, helping employees access internal information more quickly
- Engineered validation framework with 12+ automated checks reducing dataset errors by 78%, improving chatbot accuracy so internal teams could get reliable company information faster
- Designed MCP configuration architecture using JSON schemas to control chatbot behavior and response patterns across 8 prompt categories, improving response consistency by 45% through structured data governance

Codify | *Software Developer*

Sep. 2023 - Sep. 2024

- Optimized Intuit homepage performance by identifying and removing 15+ unused script dependencies and refactoring render-blocking JavaScript code, reducing initial page load time from ~10 seconds to ~5 seconds
- Delivered frontend implementations for 3 enterprise client projects including Intuit, writing 2,000+ lines of production HTML/CSS/JavaScript code that passed client acceptance testing

PROJECTS

TrendFlow | *Python, PostgreSQL, Apache Airflow, NLP, Streamlit*

- Built distributed real-time data analytics pipeline and automation tools processing 10,000+ social media posts hourly using AWS Lambda and EventBridge, predicting viral trends 24–48 hours ahead
- Optimized database performance from 8 seconds to under 1 second through normalized PostgreSQL schema design and strategic indexing, enabling responsive real-time analytics for trend tracking
- Achieved 99.5% uptime through robust error handling, automated failover, and comprehensive system monitoring

GlobalNews Translation | *Python, JavaScript, Flask, NLP, Seq2Seq, Pandas*

- Developed a multilingual seq2seq translation model for Japanese-Chinese-English, achieving 85% validation accuracy on 100,000+ sentence pairs, supporting real-time translation across three language combinations
- Deployed Flask REST API serving real-time translations with less than 200ms latency and engineered bilingual training dataset using pandas NLP techniques, improving data quality by 40% through systematic filtering

SKILLS

- **Programming & Tools:** Python, SQL, Java, JavaScript, TypeScript, HTML/CSS, Git, CI/CD
- **Technologies:** AWS (Lambda, RDS, S3, EventBridge, CloudWatch), PostgreSQL, MongoDB, DynamoDB, Django, Flask, Apache Airflow, NLP, Pandas, NumPy, REST APIs, QuickSight, Data Structures & Algorithms, Data Analytics
- **Languages:** English, Chinese, Japanese